

Customer Risk Rating - Business Requirements									
CRR Capability	BRD Ref #	Business Requirement	Additional Details (Sub-Requirements)	Purpose	Outcome	Reviewed [Y / N / WIP]	MVP (to be filled in by Product)	Release (to be filled in by Product)	
CRR Structure and Setup	12.1	System should have the ability to incorporate the core components of the CRR framework	12.1.1 - The system should have the below components that together form the CRR framework: - The CRR should have the below Risk Categories - Customer, - Geography, - Transaction, - Product, - Events 12.1.2 - Risk Elements will be categorized within the Risk Category The Risk Element will have a Weighting The Risk Element will have rules/ruleset to set the Multiplier Risk Points are the Weighting * Multiplier 12.1.3 - The multiplier of a risk element is set based on rule criteria. The system should have ability to have multiple rulesets. The system should have ability to combine data points to establish criteria. For each element users should be able to identify minimum or maximum multiplier For example, minimum for Length of Relationship and Acquisition Channel Maximum is set for all others. Potentially this is the default. System should have the ability to setup rule to trigger based on fundamental assessment risk score (1 to 10) / Centralized Lists For example, in 7,8,9 for high risk geographies System should have the ability to setup a default multiplier (to be leveraged when data is unavailable) 12.1.4 - The overall risk is determined by summing all Risk Points A threshold will be used to assign the corresponding risk rating 12.1.5 - All the specific elements in production within the current application must be available on the new platform.	Establishing the core components of the CRR framework	Ensuring current core structure and conceptual framework for deriving the CRR is in place in the POA state	Y			
CRR Change Request	12.2	System should be capable of integrating the process of change request tracking and monitoring into the platform	12.2.1 - System should have capability to facilitate requests for CRR updates 12.2.2 - The access to the Change Request UI and the capabilities within (submit/review/disposition) should be based on permissions. 12.2.3 - For the default setup, only MRO teams should be able to submit a request. Only the core CRR team (Tommy's team) should be able to review and disposition the request. 12.2.4 - The entitled user should be able to access the request page and be able to select enterprise, specific market/center(s) or legal entity and view the corresponding CRR framework (all the elements/categories and weighting) and the Fundamental Assessments/Lists. 12.2.5 - The user should be able to tag one or more attributes and indicate the change they are requesting to those attributes. Additionally, they must have an option to put in the request as a free form text (this would be necessary as an example when/if the request does not directly map to an existing risk element). 12.2.6 - They should have an option to set/select the following parameters: > Applicable 'Market/Center', 'Legal Entity', 'Product' for the change (one or more) > Type of change including rationale and driver (e.g., regulatory, others - comments being mandatory for others) 12.2.7 - Once a request is submitted, system must send a notification to the CRR team. 12.2.8 - The CRR team must be able review the request and either send it back to the submitter for clarifications or be able to update the status of the request as appropriate. (Note - CRR team will track the prioritization, tech reach out for build/effort assessments, approvals and moving change into production offline) 12.2.9 - The submitting team/user should be able to cancel/withdraw the request. A notification must be sent out to the CRR team when this occurs. 12.2.10 - The CRR team must be able to set the following statuses to the request - "Under Review", "Implemented". A notification must be sent out to the submission team/user when this occurs. 12.2.11 - All change requests and the underlying actions taken for each must be logged for audit purposes. Note: This would be a Medium priority item (not for immediate consideration). Based on a review of the new TMRPO process, this requirement maybe updated/de-scoped.	Provide the ability to systemically request for market changes and manage it via the platform	Enhancing the process to improve efficiency in the way markets place ongoing customization requests and the way in which they are tracked and implemented (~10 / month)	Y			
CRR Updates	12.3	System should enable users to make enhancements to the CRR framework	12.3.1 - System should have capability of setting up new Risk Elements / Data Points and weighting to the CRR framework (should be a configurable feature; users should be able to set this up without going through a tech development cycle). 12.3.2 - The ability to make updates should be based on user permissions (configurable). 12.3.3 - System should allow for combining multiple data points to create risk elements (example Country + Industry) > User must be able to combine two or more of the available data points to create/update a risk element. > System should throw a warning immediately if there is a constraint in creating or updating the risk element per the above. 12.3.4 - System should have the capability of tagging an element as being at Enterprise level or Local level 12.3.5 - Enterprise Risk Elements should be applicable to ALL CRR assessments (across all markets), i.e., the update made once at the enterprise level should flow down to all markets assessments. 12.3.6 - The Risk Element weighting should be at the enterprise level by default with the ability to have a local override (at the below levels) > Enterprise > Center specific [refers to the Markets / with unique combinations e.g., "International Products - D2"] > Legal Entity > Product 12.3.7 - Local Risk Elements: System should have the capability to create and identify a Risk Element as limited to certain centers/legal entities (Local element could span one or more markets/legal entities) > Local Risk Elements will only apply to the markets that it has been tagged to. 12.3.8 - PayFac, Intermediaries, GNS, etc., - for these types of products, system must provide the ability to customize the CRR framework, need to ensure that system has the data to risk rate these specific products	Redesign the way updates are made to the framework and cascaded globally	Improving the mechanics of how an update to the CRR framework is implemented globally - drastically increasing productivity and reducing the chance of manual errors	Y			
Risk Scoring - Customer Level	12.4	System should calculate the Risk Rating for each unique Customer ID	12.4.1 - The risk rating should be calculated at the Customer level 1. Consumers: Each unique Customer ID (for e.g., "C360 Customer ID" in use currently) should have a risk rating. All accounts and sub accounts within the hierarchy of this Customer ID should be pulled in for calculation of the CRR. 2. Business: Each unique Customer ID (for e.g., "Obligor ID" in use currently) should have a risk rating. All accounts and sub accounts within the hierarchy of this business entity will need to be pulled in for calculation of the CRR. 3. Risk Rating at the Amex Legal Entity level - Each unique Customer ID that spans multiple legal entities should have a CRR calculated at the legal entity level (this will be required to meet regulatory obligations). > Market consideration: The above should also apply to EMEA legal entities (AEESA, AEP parent child relationship) 12.4.2 - If any interfacing/downstream application requires account level risk rating, the system must leverage the Customer level risk rating. 12.4.3 - Users must be able to override the risk rating for a customer or a target population (from current rating to Low/Medium/High/Very High) > System should require a justification (mandatory) > Users should be able to directly set the target population via an excel upload of customers identified (template would be finalized as part of the PRD). > The ability to override would be permission based (configurable). 12.4.4 - System must have a mapping between the Risk Rating (1-10 scale normalized from the Score) and the Risk Class (Low, Medium, High) 12.4.5 - System should be able to set CRR to "Very High" for the below scenarios (respectively of the calculated points/class). The rationale for the override should be set by the system based on the scenario. 1. If an intermediary partner is in a country with a very high geography corruption risk. 12.4.6 - System should be able to set CRR to "Prohibited" for the below scenarios (respectively of the calculated points/class). The rationale for the override should be set by the system based on the scenario. 1. If the customer/business entity address is in a comprehensively sanctioned country (North Korea, Cuba, Iran, Syria) 2. If the business entity has a prohibited company structure	Enable the calculation of risk at the customer level	Consistent view of the customer risk in the ecosystem; taking into consideration all the variables that influence the risk assessment	Y			
Configurability	12.5	System should allow for flexibility in setting up rulesets	12.5.1 - Setting and updating Rules/ruleset: In addition to the "AND" clause used to create a ruleset, the system should be allow the below: > OR > Exclude/Except ("NOT") > Parentheses (to allow for more complicated rulesets) Example: Refer embedded snippet	Ability to plug in sophisticated rules/ruleset	Efficient and flexible system allowing for flexible ruleset implementation	Y			

Centralized lists / Fundamental Assessments/Notable Lists	12.6	System should have the capability to setup and maintain all reference lists	12.6.1 - System must have the capability to setup/update these reference lists at the below levels: > Enterprise > Center/Market specific > Legal Entity > Product For reference, Enterprise level lists at this time > Acquisition Channel, Industry, Geography, Company Structure, Product 12.6.2 - The multipliers used in the enterprise level lists for the should have the ability to have overrides at below levels: > Center/Market specific > Legal Entity 12.6.3 - System should provide the ability to setup Centralized (notable) lists in a user friendly manner (UI/UX to facilitate the update in an efficient manner). Product consideration - to provide options and elicit business feedback on the same. 12.6.4 - Access to setup a new list and/or update the lists will be controlled by User Access Permissions (should be configurable). 12.6.5 - The current approach of Fundamental Assessments being setup via a "Y/N question set" to compute the risk rating for the list values must be supported in the POA system.	Capability to setup reference lists at multiple levels and have the ability to override locally	Efficient and flexible system allowing for simpler implementation of reference lists	Y		
Audit on the CRR engine	12.7	System should be able to keep track of all updates made to the CRR framework and reference lists	12.7.1 - System must track and make available for viewing any change made to a List / Risk Element / Risk Category - weight and multiplier (add/modify/delete with who/when and any justification) 12.7.2 - Version history of the CRR module (enterprise level, market/center level and legal entity) must be maintained with a log of all the changes made to the underlying components (Lists/ Risk Element / Risk Category - weight and multiplier) along with user/implementation date details 12.7.3 - System must track and make available for viewing any change made to the Fundamental Assessment/Notable List/ reference data (add/modify/delete with who/when and any justification) 12.7.4 - System must track and make available for viewing any overrides to the Fundamental Assessment/Notable List reference data (add/modify/delete with who/when and any justification)	System capability to track all updates made to the framework (what/who/when/why)	Comprehensive tracking of all changes made to the application for audit and reporting purposes	Y		
Sandbox - Simulation functionality	12.8	System should support sandbox simulation functionality providing multiple instances and effective monitoring capabilities	12.8.1 - System should have the capability to run CRR simulations to analyze the impact of any CRR update to the risk distribution. 12.8.2 - The underlying data used in the simulation/sandbox should a production copy. 12.8.3 - Access to the Sandbox will be controlled by User Access Permissions (should be configurable). 12.8.4 - Multiple instances should be available for the Business (ability to run 4 instances in parallel). There should be ability to limit the sandbox to > Enterprise > Market/Center specific > Legal Entity > Product 12.8.5 - At the time of submitting/triggering a sandbox simulation, system should prompt the user to confirm that all the updates being made are in line with the required enhancement/change request (the elements that have been modified should be rendered/highlighted on the UI for easy reference/validation). 12.8.6 - The system must have the capability to track/display the progress of the simulation exercise. The estimated time to completion must also be displayed. 12.8.7 - System must notify pre-defined set of users on Sandbox completion or if there was an issue encountered in completing the process. 12.8.8 - The system must have capability to allow a simulation exercise to be cancelled at any point. The user will be required to confirm the cancellation. 12.8.9 - All reference data sets that are referenced by the CRR module should be available in the Sandbox environment. In particular (and as example) - Notable Lists, Centralized List/Fundamental Assessments 12.8.10 - For any sandbox/simulation exercise, the system must be able to display the details of all the components that were modified (element, weight, multiplier etc.,). 12.8.11 - If there's an update to a centralized list - the system must run the simulation enterprise wide - across centers and legal entities - to facilitate the understanding of the impact of the change at each level. 12.8.12 - System must maintain version control of all changes made. This must be available on the Sandbox UI. 12.8.13 - Updates made in the course of a simulation should be on top of the previous changes made (should not override prior updates). System must highlight the updates that are already in flight (in another sandbox) and allow the user to merge the changes (optional). 12.8.14 - Sandbox must provide the ability of updating the mapping between the Risk Score, Risk Rating (1-10 scale normalized from the Score) and the Risk Class (Low, Medium, High) to study the impact on the risk distribution	Ability to run simulations at enterprise and local level	Comprehensive view of the impact of any change on the customer population	Y		
Sandbox - Analysis of Results	12.9	System should provide the ability for users to review results from simulation exercises for better decision making and analytics	12.9.1 - System should be able to provide a graphic/visualization of the risk distribution changes due to an update. 12.9.2 - The results should include before and after values; provide ability to zoom in (filter by) from a global/enterprise view to a particular center/market, legal entity level or product level (and combinations thereof) 12.9.3 - The dashboard above as well as the underlying data corresponding to that view on the above dashboard must be available for download as pdf/spreadsheet 12.9.4 - A view-only access to the CRR sandbox results in the simulation should be available. The roles/groups that can utilize this capability should be configurable (default "ON" for MLRO and MCO team). 12.9.5 - System should have the ability to show a direct comparison of Current Production Risk Ratings vs Sandbox Ratings (using the underlying data at the same point in time). This should be available at various levels > zoom in from a global/enterprise view to a particular center/market, legal entity level or product level (and combinations thereof). This should be available for download as a spreadsheet/pdf. 12.9.6 - Users should be able to extract a list of all changes made in the course of a sandbox run. The before and after values should be included as applicable (a change to an existing element vs adding a new element). This should be available for download as a spreadsheet/pdf.	Ability to validate the results of simulations prior to production implementation	Comprehensive view of the impact of any change on the customer population	Y		
Model Governance	12.10	System should be able to support a 2 step approval process for promoting an update into Production	12.10.1 - Once the sandbox simulation has been validated, system must have capability to allow a user (based on permissions) to trigger/move the change in to production. 12.10.2 - The request to move to production must require a 2 step approval (must be distinct users).	System capability to ensure a maker-checker process to ensure changes are moved to Production with adequate oversight	Control over any update made to the CRR framework	Y		
CRR Certification	12.11	System should have the capability to facilitate CRR certification	12.11.1 - System must have the ability to allow for periodic or ad-hoc attestation of the CRR framework (enterprise/center/market/legal entity level). 12.11.2 - Entitled users (contingent on access; permissions based) must be able to extract out all updates made to the CRR framework between 2 user selected dates or "as-of" date and extract the changes / snapshot of the CRR framework at the appropriate level (enterprise, center/market, legal entity) and be able to attest to the alignment of the framework to the corresponding regulatory requirements. As part of attestation, users should be able to upload supporting documents.	Ability to report out the current framework at any level and highlight updates made	Attest to the compliance of the framework with the regulatory requirements	Y		
Integration with AML ecosystem	12.12	System should have the ability to make available the CRR rating to other onboarding/compliance modules within and outside the platform	12.12.1 - Integration with EDD and other applications: System must have the ability to make the Risk Rating available to downstream systems (KYC, AML Investigations, TM, EDD, Anomaly Index calc etc.,). 12.12.2 - If as part of the functionality of the integrating systems, the Risk Rating is allowed to be overridden then the same must be consumed by the CRR application along with the corresponding details (what/who/why/when). 12.12.3 - Onboarding related requirements: If an interfacing application requires the CRR to be generated or rescored as part of their process such as at onboarding or during a client refresh/review, the system must support the same. 1. System must be able to support Real-Time risk rating at onboarding (for new customers as well as in the case of new products for existing customers), using all the information that is available in the onboarding system as well as any that is already available in the firm (for existing customers) ahead of the customer approval. 2. The risk rating should be calculated using the entire CRR framework that is applicable with the available data 3. SLA / Performance: System must target appropriate SLAs (sub 2 seconds) in scoring the CRR real time in order to support immediate decision making. 12.12.4 - Data related to 'Alerts' should be available to the CRR scoring engine post disposition. The approach to ingesting these alerts and the applicable criteria would be defined separately for each category (EDD, SAR filing, Negative News, Sanctions etc.,). Product consideration: Once all the available data for each of the above category is documented, Business will provide feedback on the subset required for rule setup. 12.12.5 - The requirements in this category may be enhanced as other BRD chapters are built out that have a bearing on the Customer Risk Rating	System capability to ensure that the customer risk rating is made available to all the consuming applications	Holistic view of the CRR results across the AML ecosystem in the enterprise	Y		

Reporting / Dashboards	12.13	System should have capability to generate reports/dashboards to enable reviews and analytics	12.13.1 - System should have the ability to generate a report with all the Risk Categories/Elements/Weightings (Scoring Matrix) - This should be available at the following levels - > Enterprise > Center/Market specific > Product > Legal Entity - The user should be able to pull a as-of report for the CRR framework (e.g., system should allow the user to generate the above report at any level of granularity indicated above as of 12/31/2023). - The user should be able to generate a report that highlights the changes in the above framework (across the various levels) between 2 user specified dates. 12.13.2 - System should have the ability to generate a report with all the Fundamental Assessments / Centralized Lists / Notable Lists - This should be available at the following levels - > Enterprise > Center/Market specific > Product > Legal Entity - The user should be able to pull a as-of report for the above (e.g., system should allow the user to generate the above report at any level of granularity indicated above as of 12/31/2023). - The user should be able to generate a report that highlights the changes in the above list (across the various levels) between 2 user specified dates. 12.13.3 - System should have a CRR Risk Distribution dashboard - this will be housed in GCIP and will be the source of truth for all users (underlying data must be sent to LUMI for subsequent consumption by additional stakeholders). This will need to be dynamic to reflect the current risk distribution of the customer population. - There should ability for the user to configure the views by > Enterprise > Center/Market specific > Product > Legal Entity - The user should be able to pull a as-of report for the CRR distribution (e.g., system should allow the user to generate the above report at any level of granularity indicated above as of 12/31/2023).	Ability to pull out reports with data attributes related to CRR framework as well as the resultant risk distribution of the customer base	Enabling comprehensive review and analysis of the CRR framework in the context of enterprise and regulatory policies	Y		
User Interface/Experience	12.14	System should be able to render the CRR framework on the UI	12.14.1 - System should have the capability to render the current CRR framework on the UI i.e., Risk Categories, Risk Elements, Weighting, Overrides and corresponding details (incl market information) 12.14.2 - System should provide the user with the ability to select a specific market and render the framework in place for that market. Product design consideration - this view should be user friendly with a consideration for being able to do a side by side / overlay such that it's easy to understand what the deviations are from the enterprise framework. 12.14.3 - Users should be able to download the above into a pdf/spreadsheet. 12.14.4 - Ability to view should be based on user permissions (configurable). 12.14.5 - System should enable users to be able to place customization requests via a UI (cross-reference: Change Request requirements above)	Capability to have a UI to review the framework with the ability to scan across markets	Enhancing the user experience to have the framework easily accessible for review and updates	Y		
Customer level CRR view / Report	12.15	System should be able to provide the risk rating for a particular customer as well as make available the contributing factors that resulted in that rating	12.15.1 - System must have the ability to track/audit all updates that resulted in customer level risk rating changes 12.15.2 - Users should be able to view a customer's risk rating (based on permissions) and zoom in to the underlying factors (risk category, risk elements, points, overrides) that contributed to the risk rating (each factor such as Product X, or Industry Y for Account Z etc., and its contribution to the risk calculation; this should include any manual overrides with corresponding details). The objective would be to have transparency into the risk rating designation. Access to this would be based on user permissions (should be configurable). 12.15.3 - System must have the ability to generate a report (access to be based on permissions) to highlight risk rating changes to a customer (within a timeframe selected by the user) and the underlying reasons (what attribute changed for the customer/or which underlying account with before and after values/when the change was made/who made the change, system, user etc..)	Providing visibility into the factors that contributed to a particular customer's risk rating as well tracking the changes that caused any rescoring to an updated rating	Increase in productivity by reducing the time that is taken currently in researching the risk rating and the underlying factors contributing to it	Y		
Rescoring capabilities	12.16	System should have the capability to rescore the customer population based on changes to underlying data and/or the CRR framework	12.16.1 - System must be able to execute automated rescoring of the customer population based on the below frequency – > A monthly process to incorporate updates to transactional data and time based risk elements (e.g., length of relationship, date of establishment) that are not on the level of customer profile updates > A daily delta process that incorporates updates to a customer's profile 12.16.2 - System must have the ability to rescore population when any enhancements or updates are made to the CRR framework (at the time of implementation of the update)	Ability to apply both the CRR framework changes as well as changes in the underlying data to rescore the risk ratings for the customer population.	Enabling the second line to have a real time understanding of the risk distribution within the customer base.	Y		
Data Considerations	12.17	System implementation plan must solve for the data related challenges in the current state to ensure integrity in the CRR process	12.17.1 - Data mapping issues that exist in the current state must be resolved as part of the POA solutioning 12.17.2 - Target state platform must use the most current data that is available in the AMEX ecosystem for CRR calculations. 12.17.3 - Data must be made available to support the risk element ratings calculations to ensure a high level of integrity in the risk scoring process 12.17.4 - Full population data in the context of CRR - down to the account level - must be available to LUMI for analysis outside of GCIP. This should include all historical data. 12.17.5 - All data related to the CRR framework (elements, weightings etc..) and all updates made to it must be made to be available to LUMI for analysis outside of GCIP. This should include all historical data. 12.17.6 - Data that is related to "Legal Hold" (refers to the hold on a record that excludes it from any purge process per record retention requirements) must be made available so that it can be leveraged as needed for the CRR framework	Ensuring that the data that is sourced for purposes of CRR is accurate as well as exhaustive	Providing Business a high degree of confidence that the risk rating calculated by the system reflects the true risk of the customer	Y		
Performance Reporting	12.18	System must have the capability to track and report on the performance of the CRR application	12.18.1 - System must be able to generate the following reports related to the performance of the system: - Report detailing out any data feeds that were unable to be processed in the CRR application (incl. the feed/source, volume, reason for failure etc..). - Report that details out any breaks or issues noted in the processing of the CRR data. If the underlying issue relates to data quality, then details around the same should be included including attribute, volume impacted, fill rates (and any additional ones that are identified as being useful during the PRD phase). - Report from a DAM perspective to highlight the data attributes that could not be ingested/processed along with corresponding reason and supporting attributes. - Report providing details of stale data (dated information being used in lieu of more current information that is present elsewhere but not available due to system issues) if any that has been used in the CRR scoring/rescoring. - Report providing performance details in relation to real time CRR calculations at onboarding (new customers as well as new products for existing customers) - response times, SLA breaches, volumes etc. - Report providing details related to sandbox performance - actuals vs. the defined SLAs (reference: for vendor POC, this was 10 million accounts scored in 5 hours).	Ensuring that the data that is sourced for purposes of CRR is accurate as well as exhaustive	Providing Business a high degree of confidence that the risk rating calculated by the system reflects the true risk of the customer	Y		
Directional guidelines for moving from a rules- based framework to a AI/ML based model	12.19	System must consider the indicated guidelines while designing an AI/ML model based framework	12.19.1 - POA state could consider moving to a AI/ML based model given the ability of such a solution to analyze complex patterns across the data attributes influencing CRR and adapt to evolving risk factors. Below are a few considerations: - Regulatory Compliance and Transparency: A. Explainability: The solution must be interpretable, i.e., it must be able to provide a detailed explanation of how the risk rating was generated. B. Compliance Requirements: The solution must meet regulatory standards including guidelines on fairness, accountability and transparency. - Data Quality/Availability: A. The data considerations highlighted elsewhere in the BRD must be solved as a pre-requisite. Historical CRR data, customer profiles and past risk outcomes would be required to train a effective model B. Underlying data must span to cover all types of customers and risk scenarios to ensure ratings are not biased or inaccurate. - Integration: The solution must be integrated with existing applications related to customer onboarding, AML, TM etc., to ensure ongoing data ingestion. - Ongoing monitoring: The model must be continuously reviewed for performance, accuracy and fairness. Retraining/recalibrating maybe needed based on changes in customer behavior, market conditions or regulations.	Moving to an industry leading solution leveraging currently available capabilities in the market to derive a more accurate CRR	Enhanced accuracy and flexibility in deriving the CRR; provides value by enhancing risk detection, reducing false positives and adapting to emerging trends in customer behavior	Y		
Notifications	12.20	System should have the ability to track and notify stakeholders about any significant events that relate to CRR	12.20.1 - System must be able to send a notification to a pre-defined DL if there are significant events at any level (Enterprise wide, Market/Center, Product or Legal Entity level) > Fluctuation in risk distribution - any change in risk rating distribution by more than X% (configurable by risk rating) in either direction. 12.20.2 - The notification must include the level at which the change was detected along the % breach with before and after values. It must also indicate the underlying reason for the breach (e.g., if a certain attribute was not populated at all or was at a different fill rate than usual).	Real time notification whenever a significant event occurs in the CRR function (risk distribution changes)	Enabling the Business to be informed of critical issues immediately and respond appropriately	Y		



User Access	12.21	System should limit access to view, update and download functions based on permissions	12.21.1 User access to various functions highlighted above will be permissions based. The specific groups/users that will have access to these will be finalized as part of the PRD.	Enabling need-based access to view, update or download capability in the context of CRR related data	Ensuring adequate controls over CRR framework and related data	Y		
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